Differentiability, Limits, and Tangents

Zhixin College · Calculus I midterm review

- Differentiability: difference quotients, corners, one-sided derivatives
- Limits: one division, recognition as a derivative, one-sided sampling
- Explicit derivatives: simplify, then substitute; parity gives 0
- Implicit equations: related rates and tangent lines

Four tests

- **Differentiability.** $|x|\, g(x)$ is differentiable at 0 when $g(0) = 0$.
- **Limits.** Substitute a special function, or recognize a derivative.
- **Explicit derivatives.** At $x = 0$, read the order off the parity.
- **Implicit equations.** Substitute the point into the original equation, then differentiate.

Definition of the derivative

f is differentiable at a when the following limit exists and is finite. The two one-sided limits must agree.

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

- Differentiability implies continuity.
- Continuity only joins the function values. A corner may remain.

An absolute value times a function

At $x = 0$ the difference quotient is a sign times g:

$$\frac{|h|\, g(h)}{h} = \operatorname{sgn}(h)\, g(h).$$

- If $g(0) = 0$ and g is differentiable, the limit is 0.
- If $g(0) \neq 0$, the one-sided derivatives differ by a sign.
- $|x| \sin x$ is differentiable at 0.
- $|x|\sqrt{\sin x + 2}$ is not: the square root is about $\sqrt{2}$ at 0.

Three tests that are not differentiability

Each claim holds for $f(x) = |x|$ at 0, where f is not differentiable.

- $f(h) + f(-h) - 2f(0) \to 0$: this is continuity.
- The symmetric quotient equals 0: the one-sided slopes may oppose.
- The increment h^2 approaches from the right only.

The increment h^3 covers both sides of 0, so

$$\lim_{h \to 0} \frac{f(a + h^3) - f(a)}{h^3}$$

exists if and only if $f'(a)$ exists.

One-sided derivatives of opposite sign

$\cos|x| = \cos x$, so it is differentiable at 0. For $\cos\sqrt{|x|}$,

$$\frac{\cos\sqrt{|h|}-1}{h} \sim -\frac{\operatorname{sgn}(h)}{2}.$$

- The right-hand limit is $-\frac{1}{2}$, and the left-hand limit is $\frac{1}{2}$.
- For $|x|\sin|x|$ and $|x|\sin\sqrt{|x|}$, the extra factor forces the quotient to 0.

The derivative stops at order one

Let $f(x) = |x| \sin x$.

- $f'(0) = 0$, and f' is continuous at 0.
- The difference quotient of f' has right-hand limit 1 and left-hand limit -1.
- $f''(0)$ does not exist. The highest order is 1.

For $x \neq 0$,

$$f'(x) = \sin|x| + |x| \cos x.$$

Substitute one function

What the item contains	Substitute
A limit involving $f'(0)$	$f(x) = x$
A test that looks like differentiability	$f(x) = \lvert x \rvert$ at 0
A hypothesis on $\lvert f \rvert$ only	the constants 1 and -1, then sgn
A composition, then a limit	the jump 0 on the left and 1 on the right

Divide a rational function once

The numerator has degree one higher than the denominator:

$$\frac{x^2+1}{x+1} = x - 1 + \frac{2}{x+1}.$$

- The coefficient of x gives $a = 1$.
- The constant term gives $b = -3/2$.
- The remaining fraction tends to 0 and does not enter a or b.

The limit is already a derivative

The quotient is the derivative of $\sqrt{x^2+1}$ at $x=a$:

$$\left(\sqrt{x^2+1}\right)' = \frac{x}{\sqrt{x^2+1}}.$$

Substitution produces

$$\frac{a}{\sqrt{a^2+1}}.$$

Rationalizing cancels $x-a$ and gives the same value.

Only the absolute value has a limit

$\lim |f| = L$ fixes the absolute value.

- $f \equiv 1$ tends to L.
- $f \equiv -1$ tends to $-L$.
- The sign function satisfies $|f| \to 1$, while f itself has no limit.

None of the three specific claims holds for every such f.

A square samples only the right

x^2 approaches 0 through positive values, so $f(x^2)$ uses only the right-hand limit of f.

The next three expressions see both sides:

$$f(x^3), \qquad \big(f(x)\big)^2, \qquad f(x) - f(-x).$$

- Jump: $f = 0$ for $x < 0$ and $f = 1$ for $x \geq 0$.
- Then only $\lim f(x^2)$ exists.

As $x \to 0$,

$$1 - \cos x \sim \frac{x^2}{2}, \qquad \sin 2x \sim 2x.$$

After rationalizing, $\sqrt{3} + \sqrt{2 + \cos x} \to 2\sqrt{3}$. The quotient becomes

$$\frac{x^2/2}{2\sqrt{3} \cdot (2x)^2} = \frac{1}{16\sqrt{3}} = \frac{\sqrt{3}}{48}.$$

Simplify, then substitute

$$\left(\frac{1-\sin x}{1+\sin x}\right)' = \frac{-2\cos x}{(1+\sin x)^2}.$$

At $x=\pi/6$ the value is $-4\sqrt{3}/9$. The form $-4/(3\sqrt{3})$ is the same number.

- A product splits into two terms. For $\sqrt{x}\sin x$, differentiate each factor once.
- If only one point is asked, evaluate each factor and each derivative at that point.

Parity gives the value at zero

- An even-order derivative of an odd function is odd.
- An odd-order derivative of an even function is odd.
- An odd function, where defined at 0, equals 0.
- $\tan x$ is odd, so its fourth derivative at 0 is 0.
- $1/(1+x^2)$ is even, so its third derivative at 0 is 0.

Implicit differentiation: substitute first

1. Substitute the given x and read off y or f.
2. Differentiate both sides.
3. Substitute the point again. A factor of x vanishes at $x = 0$.

For $f(x) + x\sin(f(x)) = x^2 + 1$, one has $f(0) = 1$ and

$$f'(0) + \sin 1 = 0, \qquad f'(0) = -\sin 1.$$

Differentiate a second time

From $y^3 + y = 2\cos x$, the left side is strictly increasing in y, so $x = 0$ forces $y = 1$. One differentiation gives $y'(0) = 0$.

The term $(y')^2$ vanishes at this point. The second derivative satisfies

$$(3y^2 + 1)y'' = -2\cos x.$$

At $y = 1$ and $x = 0$,

$$y''(0) = -\tfrac{1}{2}.$$

Differentiate the constraint in t. Each variable carries its own rate:

$$xx' + yy' + zz' = 0.$$

- The point $(3, 4, 12)$ lies on the sphere: $9 + 16 + 144 = 169$.
- With $x' = 4$ and $y' = 3$,

$$12 + 12 + 12z' = 0, \qquad z' = -2.$$

Tangency means equal slopes

Set the derivative of the curve equal to the slope of the line.

- $3x^2 - 3 = 9$ gives $x = \pm 2$. Both points are required.
- $x = 2$ gives $b = -16$; $x = -2$ gives $b = 16$.

On $x^2 - y^2 = 9$, one has $y' = x/y$. At $(5, -4)$ the slope is $-5/4$, so the tangent line is

$$5x + 4y - 9 = 0.$$

If you see	Do this first
A difference quotient	Try $f(x) = x$ or $\lvert x\rvert$
Degree one higher	Divide once; match coefficients
Square roots over $x - a$	Recognize a derivative, or rationalize
A limit of $\lvert f\rvert$ only	Constants 1, -1, then sgn
$f(x^2)$ or $f(x^3)$	A jump; see which side is used
A high derivative at 0	Use parity
An implicit equation or a tangent	Substitute the point; equate slopes

Stock examples

Function	**What it decides**
$\lvert x\rvert$ at 0	Symmetric quotient, two-sided sum, right-only increment
$f(x) = x$	A limit that contains $f'(0)$
Constants 1 and -1	Separates the limits L and $-L$
Sign function	$\lvert f\rvert$ may have a limit while f does not
Jump, 0 then 1	$f(x^2)$ has a limit; $f(x^3)$ does not
$\lvert x\rvert \sin x$	Differentiable once; $f''(0)$ fails
Odd and even functions	Odd function, even order: 0. Even function, odd order: 0.

Three lines on the paper

1. Write the answer: the letter, or the final number.
2. Name the test or the example, and record what the substitution gives.
3. Finish the calculation: the division, the quotient, or the differentiation, through the substitution.

If a tangent condition has two points of contact, record both values of b.